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ABSTRACT

Road accidents involving two-wheeler riders often lead to severe injuries due to delayed emergency response and poor helmet compliance. This project presents a Smart Helmet System designed to enhance rider safety using embedded and wireless technologies.

The system is built around an **ESP32** microcontroller and integrates an **MPU-6050** accelerometer and gyroscope to detect sudden impacts indicating an accident. Upon crash detection, a **NEO-6M GPS** module acquires the rider's real-time location, which is transmitted as an emergency SMS to predefined contacts through a **SIM800L GSM** module. A helmet-wear detection mechanism is implemented to ensure that the motorcycle ignition is enabled only when the helmet is properly worn using a relay-based interlock system.

Additionally, Bluetooth connectivity supports hands-free operation, reducing rider distraction. The proposed system is low-cost, compact, and reliable, making it suitable for academic demonstration and practical road safety applications.

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THANK YOU